

Cryptographic Electronics

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Abstract—Cryptographic Electronics proposes devices to enforce agreed conditions for trust, protect sensitive assets, and help parties coordinate without surrendering the secrets behind their strategic advantage. We seek funding, engineering collaborators, and prototyping resources to develop these ideas through hardware synthesis and FPGA evaluation. The opportunity is to make cryptographic protections practical where people and systems interact. The program is at the concept and design stage; support would enable prototypes and evaluation with prospective users.

Index Terms—Cryptographic devices, hardware synthesis, FPGA.

I. THE NEED AND OPPORTUNITY

Cooperation should not require surrendering the information that makes a person or organization competitive. Parties may need to coordinate while protecting proprietary knowledge, credentials, and control over their actions. Our idea is to make the conditions for trust enforceable through devices that carry out agreed rules and protect the secrets entrusted to them. Secure multiparty computation and cryptographic approaches to mediated games offer foundations for private coordination [1], [2]; each application must specify what its outputs reveal and the assumptions its protections require.

Prospective customers include businesses collaborating across organizational boundaries, operators protecting valuable credentials, and institutions relying on independently verifiable records. The product opportunity is to give these protections an inspectable physical boundary, reducing dependence on a counterparty or shared administrator. Pilot users would help test that value against integration effort and cost, and establish which needs justify a dedicated device.

Examples include cameras and input peripherals, USB login keys, acoustic devices such as WaveSign, and Waajacu Data Diode and Vault. These illustrate possible applications; prototype selection and detailed architectures remain open.

II. FROM CRYPTOGRAPHY TO HARDWARE

Hardware synthesis would connect this purpose to physical prototypes. Precise interfaces and executable reference models would guide synthesizable register-transfer-level (RTL) descriptions in Verilog or VHDL. Simulation, synthesis, placement and routing, and FPGA experiments would establish their behavior and cost. Hardware and software responsibilities would follow the application, using established cryptographic implementations and components where appropriate.

Reference comparisons and suitable formal checks would assess whether implementations follow their specifications.

Security evaluation would separately examine key handling, failure behavior, exposed interfaces, and relevant leakage or fault attacks. Synthesis alone does not establish the intended protections.

III. PROTOTYPE DEVELOPMENT

The first supported stage would identify a user need, select a bounded application, and define its threat model and acceptance criteria. Subsequent work would produce reference software, test vectors, synthesizable modules, FPGA builds, and the physical integration needed for a demonstrator. Independent review would help assess what the results establish and what remains unresolved.

Experiments would measure resource use, latency, throughput, and energy consumption, alongside usability and integration effort in the intended workflow. Tests would cover ordinary operation, malformed inputs, and failure conditions. Documented hardware, tools, workloads, and measurement conditions would make the results reproducible.

The resulting design sources, documented build, verification materials, and measurements would form a foundation for further devices. Platform, schedule, and budget would follow the agreed prototype scope.

IV. SUPPORT AND COLLABORATION

We seek financial support and collaborators to turn these ideas into hardware prototypes. Funding would support dedicated engineering time, cryptographic and protocol review, RTL development, verification, board fabrication and assembly, integration, and experimental evaluation. FPGA development boards, synthesis and verification tools, fabrication services, and access to measurement equipment or laboratories would also make direct contributions to the work.

We welcome collaboration in applied cryptography, digital design and synthesis, embedded systems, PCB engineering, formal methods, and hardware security evaluation. Partners with practical use cases can help identify which prototype would be most useful and what evidence would justify further development.

REFERENCES

- [1] O. Goldreich, S. Micali, and A. Wigderson, “How to play ANY mental game,” in *Proc. 19th Annual ACM Symposium on Theory of Computing (STOC)*, May 1987, pp. 218–229, doi:10.1145/28395.28420.
- [2] Y. Dodis, S. Halevi, and T. Rabin, “A cryptographic solution to a game theoretic problem,” in *Advances in Cryptology—CRYPTO 2000*, ser. Lecture Notes in Computer Science, vol. 1880, 2000, pp. 112–130, doi:10.1007/3-540-44598-6_7.